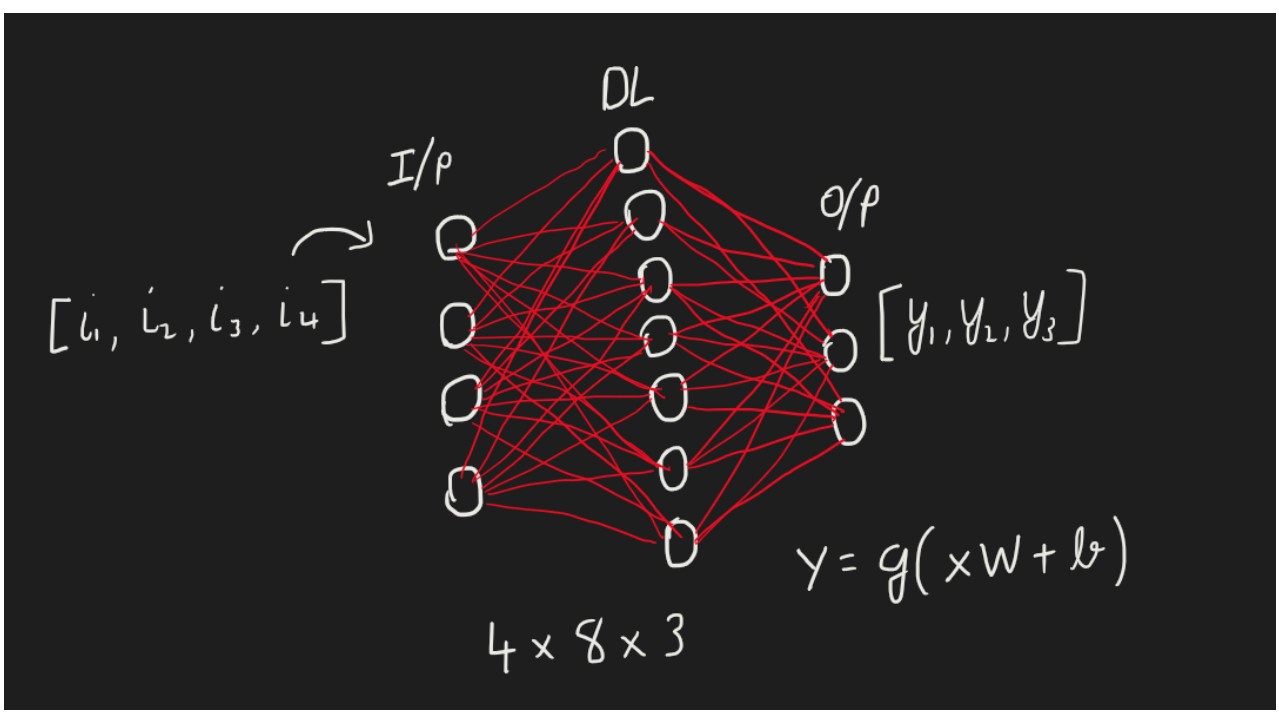
EEA EndTerm-Report
Neural Network in FPGA's:

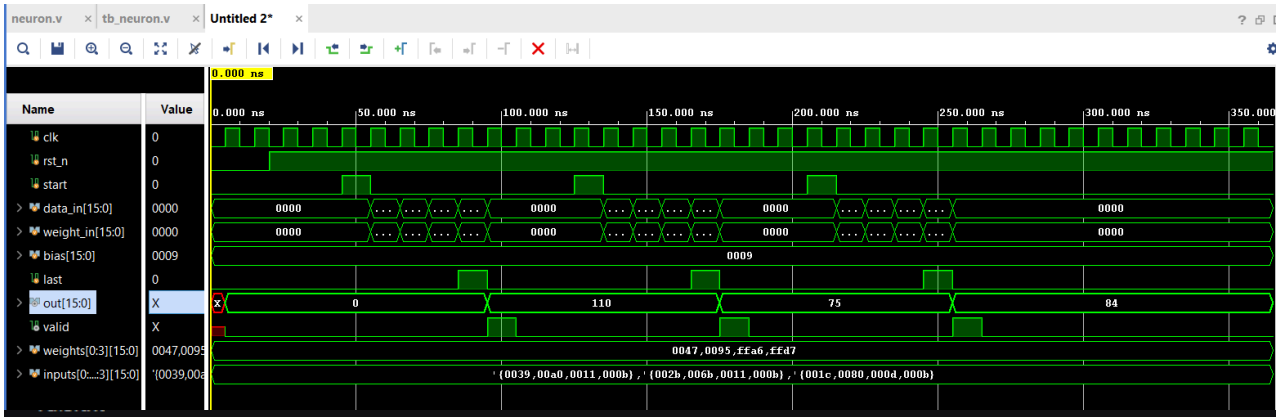
Objective:

To design and implement a simple multilayered Neural Network on FPGAs using verilog to increase computation speed by harnessing parallelism.

Approach:

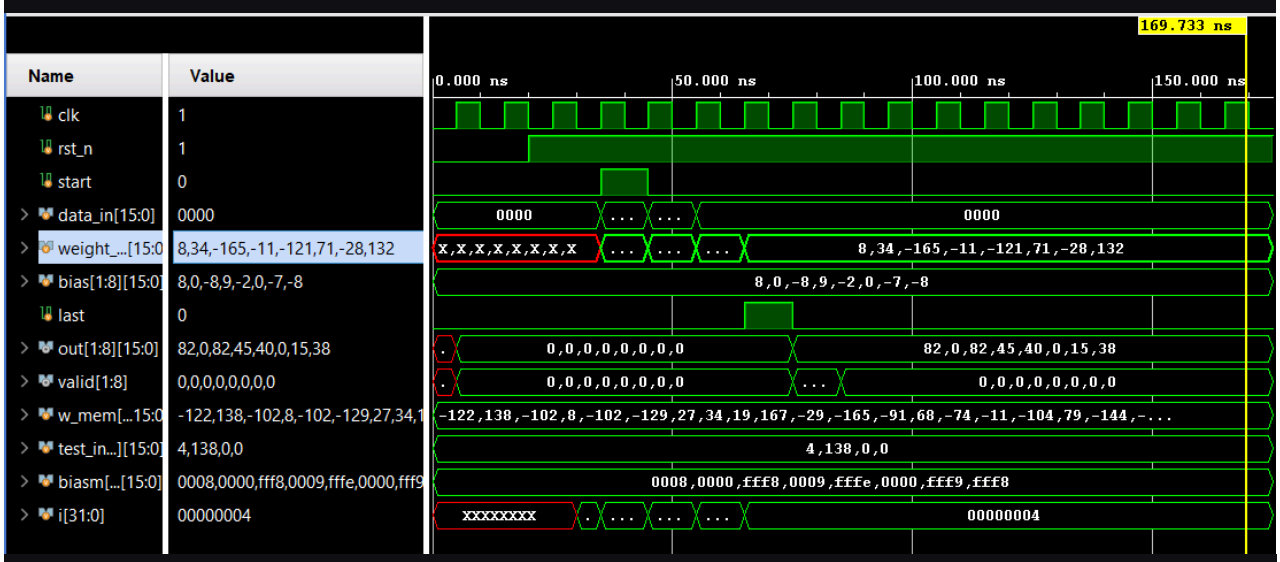
Built a sequential neuron component in verilog using MAC operation to accumulate input weight product summed with bias and applied ReLU for final activation output, Built a deep layer by combining 8 neurons with equivalent inputs into elements for parallel computation and utilised BRAM's for weight & bias storage, Trained a tiny model (4 inp, 8 dense, 3 out) layers on IRIS dataset and normalised to obtain informations in Q8 format, designed nntop to connect the layer to produce obtained output and evaluated/optimised the I/O connections to fit the given board with total LUT: 624 and a total of 380 FF's With WNS:-0.35, TNS:-5.5 and a total power consumption of 0.1772. Below shows the timing diagram of Neuron and Layer module and their respective input along with the schematic of the network implemented.






```
    input vector/list                                          output verctor/list
1:  [0.22222222 0.625      0.06779661 0.04166667]              0.43027764 ~ 110
2:  [0.16666667 0.41666667 0.06779661 0.04166667]              0.29401153 ~ 75
3:  [0.11111111 0.5        0.05084746 0.04166667]              0.33298856 ~ 84


neuron weights: [ 0.2763999 ,  0.5803707 , -0.3521333 , -0.16179001]
uint16                [71, 149, -90, -41]
neuron bias:  0.03673845     = 9
```

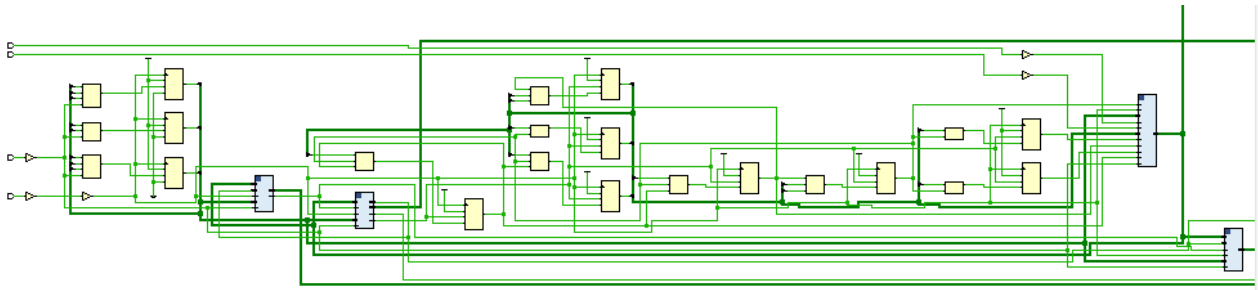



```
8 layer dense stuff:
weight vec:
    input to dense 8 : [1.3203894, 1.1952475, 0.21684133, 0.0]     [338, 306, 55, 0]
    [[-122, -102,   19,  -91, -104,   20,    8, -122],
      [ 138, -129,  167,   68,   79,  -58,   41,   87],
      [-102,   27,  -29,  -74, -144,  -32, -165, -110],
      [   8,   34, -165,  -11, -121,   71,  -28,  132]]
```

Results Obtained from Implementation:

WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP
											624	**380**	**0**	**0**	**6**
-0.349	-5.546	0.210	0.000		0.000	0.172	0	71 Warn			612	382	0	0	6

The obtained results suggest that given clock frequency would be slow for the task since WNS is -ve hence it would be better to increase clock frequency, LUT utilisation could be brought down drastically by proper utilisation of BRAM for data storage.



Personally the hardest part was the implementation of nntop since it required optimisation of n.o I/O pins for my board(Basys3) and using BRAM's to store serialised weights and biases data since readmemb doesn't read pickled binary files in the same format. Sequentially passing in the input was tricky at times, due to improper usage of blocking assignment, Double check the structure of .mem files such that proper values could be attained with test cases.

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